Anomaly Identification and Detection Using Statistical and Machine Learning Methods

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By

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Abstract

This study investigates methods for identifying anomalies in retail demand forecasting data using both statistical techniques and machine learning models. The objective is to detect outliers in daily production forecasts to reduce forecast error in perishable food inventory planning. Leveraging a continuously expanding large scale dataset with records spanning a number of years up to the current date, the project utilizes a cloud-based data platform to develop an end-to-end anomaly detection pipeline, including data cleaning, statistical analysis, and model development.

Anomalies are defined as forecast errors exceeding ± 3 standard deviations from the mean. To enhance performance, the data are segmented by item-store combinations, and feature engineering techniques are applied to generate variables that improve separability between normal and anomalous observations. An Isolation Forest model is then trained, selected for its effectiveness in identifying anomalies in highly imbalanced datasets. Model thresholds are adjusted to prioritize the detection of the minority class and minimize missed outliers.

The results demonstrate the value of combining statistical rules with machine learning to create a scalable and robust framework for identifying errors in perishable item demand forecasting within a large retail environment.